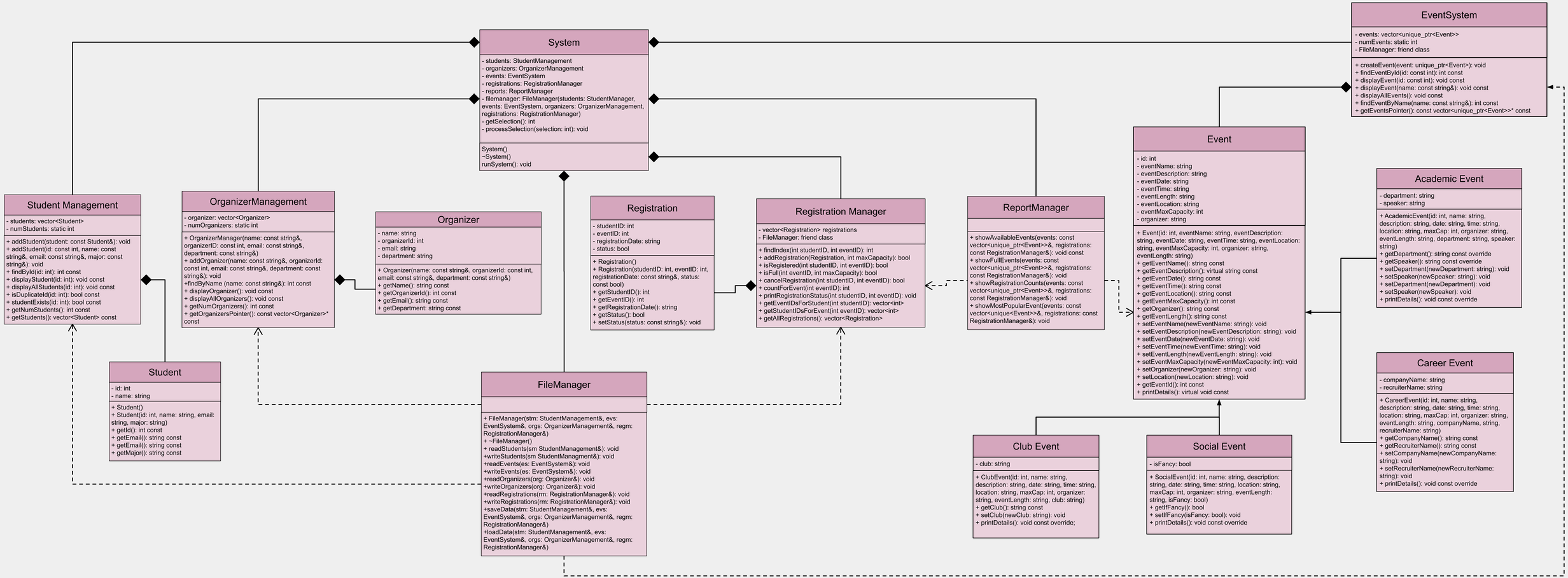
System
- students: StudentManagement
- organizers: OrganizerManagement
- events: EventSystem
- registrations: RegistrationManager
- reports: ReportManager
- filemanager: FileManager(students: StudentManager, events: EventSystem, organizers: OrganizerManagement, registrations: RegistrationManager)
- getSelection(): int
- processSelection(selection: int): void
System()
~System()
runSystem(): void
EventSystem
- events: vector<unique_ptr<Event>>
- numEvents: static int
- FileManager: friend class
+ createEvent(event: unique_ptr<Event>): void
+ findEventById(id: const int): int const
+ displayEvent(id: const int): void const
+ displayEvent(name: const string&): void const
+ displayAllEvents(): void const
+ findEventByName(name: const string&): int const
+ getEventsPointer(): const vector<unique_ptr<Event>>* const
Student Management
- students: vector<Student>
- numStudents: static int
+ addStudent(student: const Student&): void
+ addStudent(id: const int, name: const string&, email: const string&, major: const string&): void
+ findById(id: int): int const
+ displayStudent(id: int): void const
+ displayAllStudents(id: int): void const
+ isDuplicateId(id: int): bool const
+ studentExists(id: int): bool const
+ getNumStudents(): int const
+ getStudents(): vector<Student> const
OrganizerManagement
- organizer: vector<Organizer>
- numOrganizers: static int
+ OrganizerManager(name: const string&, organizerID: const int, email: const string&, department: const string&)
+ addOrganizer(name: const string&, organizerId: const int, email: const string&, department: const string&): void
+findByName (name: const string&): int const
+ displayOrganizer(): void const
+ displayAllOrganizers(): void const
+ getNumOrganizers(): int const
+ getOrganizersPointer(): const vector<Organizer>* const
Organizer
- name: string
- organizerId: int
- email: string
- department: string
+ Organizer(name: const string&, organizerId: const int, email: const string&, department: const string&)
+ getName(): string const
+ getOrganizerId(): int const
+ getEmail(): string const
+ getDepartment: string const
Registration
- studentID: int
- eventID: int
- registrationDate: string
- status: bool
+ Registration()
+ Registration(studentID: int, eventID: int, registrationDate: const string&, status: const bool)
+ getStudentID(): int
+ getEventID(): int
+ getRegistrationDate(): string
+ getStatus(): bool
+ setStatus(status: const string&): void
Registration Manager
- vector<Registration> registrations
- FileManager: friend class
+ findIndex(int studentID, int eventID): int
+ addRegistration(Registration, int maxCapacity): bool
+ isRegistered(int studentID, int eventID): bool
+ isFull(int eventID, int maxCapacity): bool
+ cancelRegistration(int studentID, int eventID): bool
+ countForEvent(int eventID): int
+ printRegistrationStatus(int studentID, int eventID): void
+ getEventIDsForStudent(int studentID): vector<int>
+ getStudentIDsForEvent(int eventID): vector<int>
+ getAllRegistrations(): vector<Registration>
ReportManager
+ showAvailableEvents(events: const vector<unique_ptr<Event>>&, registrations: const RegistrationManager&): void const
+ showFullEvents(events: const vector<unique_ptr<Event>>&, registrations: const RegistrationManager&): void
+ showRegistrationCounts(events: const vector<unique_ptr<Event>>&, registrations: const RegistrationManager&): void
+ showMostPopularEvent(events: const vector<unique<Event>>&, registrations: const RegistrationManager&): void
Event
- id: int
- eventName: string
- eventDescription: string
- eventDate: string
- eventTime: string
- eventLength: string
- eventLocation: string
- eventMaxCapacity: int
- organizer: string
+ Event(id: int, eventName: string, eventDescription: string, eventDate: string, eventTime: string, eventLocation: string, eventMaxCapacity: int, organizer: string, eventLength: string)
+ getEventName(): string const
+ getEventDescription(): virtual string const
+ getEventDate(): string const
+ getEventTime(): string const
+ getEventLocation(): string const
+ getEventMaxCapacity(): int const
+ getOrganizer(): string const
+ getEventLength(): string const
+ setEventName(newEventName: string): void
+ setEventDescription(newEventDescription: string): void
+ setEventDate(newEventDate: string): void
+ setEventTime(newEventTime: string): void
+ setEventLength(newEventLength: string): void
+ setEventMaxCapacity(newEventMaxCapacity: int): void
+ setOrganizer(newOrganizer: string): void
+ setLocation(newLocation: string): void
+ getEventId(): int const
+ printDetails(): virtual void const
Academic Event
- department: string
- speaker: string
+ AcademicEvent(id: int, name: string, description: string, date: string, time: string, location: string, maxCap: int, organizer: string, eventLength: string, department: string, speaker: string)
+ getDepartment(): string const override
+ getSpeaker(): string const override
+ setDepartment(newDepartment: string): void
+ setSpeaker(newSpeaker: string): void
+ setDepartment(newDepartment): void
+ setSpeaker(newSpeaker): void
+ printDetails(): void const override
Career Event
- companyName: string
- recruiterName: string
+ CareerEvent(id: int, name: string, description: string, date: string, time: string, location: string, maxCap: int, organizer: string, eventLength: string, companyName, string, recruiterName: string)
+ getCompanyName(): string const
+ getRecruiterName(): string const
+ setCompanyName(newCompanyName: string): void
+ setRecruiterName(newRecruiterName: string): void
+ printDetails(): void const override
Student
- id: int
- name: string
+ Student()
+ Student(id: int, name: string, email: string, major: string)
+ getId(): int const
+ getEmail(): string const
+ getEmail(): string const
+ getMajor(): string const
FileManager
+ FileManager(stm: StudentManagement&, evs: EventSystem&, orgs: OrganizerManagement&, regm: RegistrationManager&)
+ ~FileManager()
+ readStudents(sm StudentManagement&): void
+writeStudents(sm StudentManagment&): void
+readEvents(es: EventSystem&): void
+writeEvents(es: EventSystem&): void
+readOrganizers(org: Organizer&): void
+writeOrganizers(org: Organizer&): void
+readRegistrations(rm: RegistrationManager&): void
+writeRegistrations(rm: RegistrationManager&): void
+saveData(stm: StudentManagement&, evs: EventSystem&, orgs: OrganizerManagement&, regm: RegistrationManager&)
+loadData(stm: StudentManagement&, evs: EventSystem&, orgs: OrganizerManagement&, regm: RegistrationManager&)
Club Event
- club: string
+ ClubEvent(id: int, name: string, description: string, date: string, time: string, location: string, maxCap: int, organizer: string, eventLength: string, club: string)
+ getClub(): string const
+ setClub(newClub: string): void
+ printDetails(): void const override;
Social Event
- isFancy: bool
+ SocialEvent(id: int, name: string, description: string, date: string, time: string, location: string, maxCap: int, organizer: string, eventLength: string, isFancy: bool)
+ getIfFancy(): bool
+ setIfFancy(isFancy: bool): void
+ printDetails(): void const override